OnTrack: System Features & Specifications

Mobile Application Technology — Technical Specifications & AI Usage Disclosure

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System Overview: OnTrack is a cross-platform mobile application and academic administration portal engineered for On-the-Job Training (OJT) monitoring. It replaces manual Daily Time Records (DTR) by integrating automated rendering hour computation, native GPS geolocation verification, a live real-time system clock, student task deliverables management, and comprehensive academic coordinator oversight.

PART 1: Client Side (Student Intern Mobile App)

The client-side mobile application is built using Ionic 9 and Angular 22 (Standalone Components) paired with Capacitor native plugins. It provides interns with an intuitive interface to record attendance, track rendering hours, log daily accomplishments, and manage deliverables.

1. Authentication & Onboarding

- **Multi-Step Student Registration:** Guided 3-step registration workflow that captures comprehensive student OJT information.
  - **Step 1 (Academic Details):** Inputs Student ID Number, Full Name, Academic Program, Year Level, and Section selection (BSIT 4-A to 4-K).
  - **Automated Coordinator Pairing:** Selecting an academic section automatically pairs the student with that section's designated faculty coordinator.
  - **Step 2 (OJT Placement):** Records Host Training Establishment (HTE) / Company Name, Company Address, Trainer Name, Start Date, Target End Date, and Required Hours.
  - **Step 3 (Security & Review):** Secure password creation with BCrypt encryption, password visibility toggles, and an instant summary review card before submitting.
- **Unified Sign-In & Session Handling:** Secure authentication using Student ID or Institutional Email with hashed password verification. Includes persistent local session storage and Angular Route Guards.

2. Student Dashboard (Home)

- **Profile Header:** Displays user avatar with initials, active duty beacon indicator, student name, student ID, academic section, and assigned company badge.
- **Daily Attendance Console:** Streamlined single-card attendance console with one-tap Time In / Time Out button, dynamic duty indicator, and shift start timestamp.

- **Rendered vs. Required Hours:** Real-time counter displaying accumulated rendering hours against the target requirement (e.g., 120.00 / 486 hrs).
- **Visual Progress Ring & Bar:** Live percentage indicator illustrating overall OJT completion progress (0% to 100%).
- **Hours Remaining Metric:** Automatically computed countdown showing exact hours left to complete internship requirements.
- **Target Completion Countdown:** Calculates calendar days remaining until the intern's target completion date.
- **Edit Goal Modal:** Allows students to adjust their required hours target (e.g., 486 or 500 hours) and targeted completion date.
- **Quick Navigation Shortcuts:** Grid providing one-tap access to Time Logs (DTR), Tasks, Reports, and Profile.
- **Recent Activity Feeds:** Instant preview of latest attendance logs and pending tasks directly on the main dashboard feed.

3. GPS Attendance & Duty Tracking (Time Logs / DTR)

- **GPS Geolocation Verification:** Captures high-accuracy latitude and longitude coordinates upon Time In via @capacitor/geolocation to verify physical workplace presence.
- **Live Realtime System Clock:** Displays a live ticking digital clock (HH:MM:SS AM/PM) with current date and pulsing live beacon to record exact check-in timing.
- **Location Name Resolution:** Resolves GPS coordinates into human-readable workplace street names and establishments with a one-tap manual refresh option.
- **Tactile Haptic Feedback:** Triggers device vibration via @capacitor/haptics upon successful Time In or Time Out.
- **Time-Out Accomplishment Logging:** Includes an accomplishment summary text area where students log daily duties and tasks finished upon duty completion.
- **Automated Hour Computation:** Accurately computes total hours rendered for the shift based on Time In and Time Out timestamps.
- **DTR History & Timesheet Explorer:** Complete reverse-chronological list of past attendance entries showing date, timestamps, rendered hours, accomplishment logs, and coordinates.

4. Task & Deliverable Management (Tasks)

- **Task Status Filtering:** Organizes deliverables into All, Pending, In Progress, and Completed filter tabs.
- **Personal Task Creation:** Interns can create deliverables with custom titles, scope descriptions, and target completion deadlines.
- **Deliverable Submission & Proof Upload:** Allows interns to upload proof files (PDFs, images) or external repository/documentation links to prove task completion.
- **Status Toggling:** One-tap mechanism to update task status from pending to completed.

5. Reports & Profile Management

- **Structured Summary Report:** Organized 3-column metadata header (Student Info, Placement, Hours & Tasks) followed by Time Logs table (Date, Time, Hours) and Tasks Overview table (Task Name, Description, Status, Due Date).
- **Digital DTR Export & Print:** Prepares clean, high-contrast attendance summary tables ready for printing and academic submission via print-ready stylesheets.
- **Profile & Contact Management:** Displays academic details, adviser credentials, and emergency contact details with secure one-tap logout.

PART 2: Admin / Coordinator Side (Academic Faculty Portal)

The coordinator portal provides academic OJT supervisors with real-time oversight tools to monitor student attendance, audit Daily Time Records, inspect GPS coordinates, review student deliverables, and filter cohorts by academic section.

1. Coordinator Authentication & Section Filtering

- **Faculty Authentication:** Dedicated faculty sign-in using Employee ID (e.g., PROF-2026-001) or institutional email with role-based permissions (admin, coordinator, adviser).
- **Interactive Section Switcher:** Header tab dropdown allowing the coordinator to instantly filter the dashboard across sections (BSIT 4-A to 4-K or All Sections).
- **Automated Roster Scoping:** Selecting a section dynamically scopes all KPI analytics, intern lists, and attendance logs to that class.

2. System-Wide Analytics & KPI Metrics

- **Total Monitored Interns:** Live count of registered students currently active in the selected academic section.
- **Active on Duty Counter:** Real-time count of students who are currently on active duty at their company placements.
- **Total Rendered Hours:** Aggregate sum of all internship hours rendered across the student cohort.
- **Task Completion Metrics:** Visual breakdown of pending versus completed student deliverables.

3. Monitored Intern Roster

- **Live Search & Filter:** Filter students in real time by full name, school ID number, or company placement.
- **Duty Status Chips:** Instant filter pills for All, On Duty (live green beacon), and Off Duty (gray indicator).
- **Intern Summary Cards:** Displays student name, student number, company agency, live duty badge, hours remaining, and progress bar (e.g., 245 / 486 hrs · 50.4%).
- **Quick View Action Button:** One-click 'View >' button on every card that opens the student's complete record.

4. Detailed Student Record & Audit View

- **Student Overview Modal:** In-depth modal presenting the student's complete profile, contact details, total hours rendered, remaining hours, and progress percentage.
  - **Tab 1: Complete Attendance History (DTR Audit):** Audit log showing every shift rendered by the student, including date, exact Time In, Time Out, and rendered hours.
  - **Daily Accomplishment Review:** Inspects the daily accomplishment notes written by the student upon time-out.
  - **GPS Coordinates Inspection:** Verifies exact GPS latitude and longitude captured at the moment of time-in.
  - **Tab 2: Student Tasks & Deliverables:** Displays all deliverables submitted by the student with status indicators and submitted attachments.

5. Student Task Monitoring & Proof Verification Console

- **Task Overview Metrics:** Summary statistics of Total Tasks, Pending Tasks, and Completed Tasks across the section.
- **Deliverables Table:** Interactive table displaying student name, task title, completion date, and file attachment badges.
- **Document & Proof Inspection:** One-click modal to preview and download submitted PDF documents, diagrams, or image proofs.
- **Official Deliverable Verification Stamp:** Allows the coordinator to officially verify and approve submitted internship milestones with a digital verification stamp and timestamp.

6. Real-Time Attendance & Verification Feed

- **Live Duty Feed:** Chronological stream of real-time check-in events across the student cohort.
- **Compliance Audit:** Enables immediate auditing of whether students arrive at their workplaces on time and within assigned geographical boundaries.

PART 3: Feature Comparison Matrix

The table below compares the functional responsibilities between the Student Intern and Academic Coordinator sides of the platform:

Capability / Functional Area	Student Intern (Client Side)	Academic Coordinator (Admin Side)
Authentication & Access	Unified login & multi-step registration	Unified login with role-based faculty permissions
Section Assignment	Selects section (BSIT 4-A to 4-K)	Switches & filters by sections (A to K)

Duty Status Tracking	Single-tap Time In / Time Out	Live monitoring of on-duty interns
GPS Verification	Captures device latitude/longitude	Audits coordinates and location tags
Realtime Clock	Live ticking system clock display	Audits exact server & client timestamps
Rendered Hours Computation	Automated daily & remaining hours	Cohort-wide aggregated hours & stats
Accomplishment Logging	Submits daily accomplishment logs	Audits accomplishment notes per shift
Task & Deliverables	Creates tasks and uploads proof files	Monitors deliverables & verifies proofs
Student Record Audit	Views personal metrics & progress	Audits complete DTR history & deliverables
Timesheet & DTR Export	Exports formatted timesheet summary	Inspects digital timesheets across weeks

PART 4: Technical Implementation

This section details the technical architecture, native Capacitor capabilities, data handling strategies, RESTful API operations, and pivotal engineering choices implemented across the OnTrack system.

1. Relevant Capacitor Capabilities & Native Hardware Integration

- **@capacitor/geolocation (Workplace GPS Verification):** Interfaces directly with Android native location services (GPS hardware and fused network positioning). Upon initiating Time-In, it captures high-accuracy latitude and longitude coordinates to verify the intern's physical presence at the designated training company. Coordinates are cached in-memory to prevent repeated satellite acquisition delays, and reverse geocoding via OpenStreetMap Nominatim translates raw coordinates into human-readable workplace street addresses.
- **@capacitor/camera (Deliverable & Profile Capture):** Bridges native mobile camera hardware and device photo albums. Allows interns to capture profile avatars and attach photographic proof of completed tasks and project deliverables. Handles runtime permissions (CAMERA and READ_MEDIA_IMAGES), automatic image orientation normalization, and Base64/Blob conversion with JPEG compression to minimize bandwidth and storage overhead.
- **@capacitor/haptics (Tactile Feedback):** Engages the physical device vibration motor to provide immediate tactile feedback upon user interactions. Triggers distinct sensory feedback patterns—such as medium impacts when opening dialogs, submitting attendance logs, or completing

deliverables, and notification success bursts upon successful backend API confirmation—enhancing the mobile user experience.

- **@capacitor/status-bar & @capacitor/keyboard (Native Viewport Polish):** Dynamically configures the native Android status bar styling to align with OnTrack's dark-navy brand aesthetic (#2B5578) with light icon themes. Simultaneously listens to native soft keyboard appearance and dismissal events, ensuring input fields automatically scroll into view without layout clipping or modal overflow.
- **@capacitor/app (Application Lifecycle Management):** Monitors native application lifecycle states (app pause, resume, and hardware back-button navigation). Ensures that active timers, background requests, and unsaved form states recover gracefully without data loss when switching apps.

2. Data Handling & State Architecture

- **Angular 22 Standalone Architecture:** Built with modern Angular standalone components without NgModules. Each page and UI component independently imports its necessary dependencies (Ionic standalone directives, CommonModule, ReactiveFormsModule), enabling granular tree-shaking, fast cold-start performance, and reduced JavaScript bundle sizes on mobile devices.
- **Reactive Form Validation & Strict Typing:** All inputs are managed through Angular Reactive Forms (FormGroup, FormBuilder) backed by TypeScript models. Multi-step forms (such as the 3-step registration wizard) maintain step-specific validation gates, prevent premature submissions, and provide real-time field error hints.
- **RxJS Observable Streams & Reactive State:** Authentication and user session states are managed using RxJS BehaviorSubjects within AuthService. State transitions automatically propagate across components, enabling live updates to user avatars, duty beacons, and rendered hour statistics without manual reloads.
- **Local Storage Persistence & Route Protection:** Authenticated user credentials, profile records, and active sessions persist in browser/device localStorage. Angular Route Guards (canActivate) inspect active session tokens and role metadata, automatically redirecting unauthenticated users to login and preventing students from accessing administrative coordinator endpoints.
- **Decoupled Real-Time Clock Engine:** A dedicated 1,000ms system timer drives the live digital clock (HH:MM:SS AM/PM) and formatted date on the Time-In console. Updates trigger Angular ChangeDetectorRef to ensure smooth second-by-second UI rendering, and the interval is cleanly torn down in ngOnDestroy() to prevent background memory leaks.
- **Location Data Caching:** Coordinates and reverse-geocoded location strings are cached in-memory with timestamp freshness markers. If the device already has a recent valid position, the modal renders immediately while fresh satellite polling executes seamlessly in the background.

3. API Operations & Backend Architecture (XAMPP / PHP 8.x + MySQL)

- **RESTful JSON Communication Standards:** All backend endpoints communicate exclusively using application/json payloads and return standardized response envelopes (success boolean, message string, and typed data payload). Standard HTTP response codes (200 OK, 400 Bad Request, 401

Unauthorized, 409 Conflict) are strictly enforced, alongside comprehensive CORS headers for frictionless cross-origin mobile communication.

- **Authentication Endpoints (/api/auth/):** POST /api/auth/register.php processes multi-step student registration, verifies email formatting, enforces unique school IDs, and automatically binds students to coordinators based on academic section. POST /api/auth/login.php provides unified authentication across students and faculty, validating passwords against BCrypt hashes (password_verify).
- **Attendance & DTR Operations (/api/attendance/):** POST /api/attendance/clock_in.php validates active duty state, records timestamp and GPS coordinates, and initializes shift status. POST /api/attendance/clock_out.php computes shift duration using MySQL TIMEDIFF, saves student accomplishment notes, and marks the shift completed. GET /api/attendance/history.php returns reverse-chronological DTR history optimized for timesheet audits.
- **Dashboard & Analytics Endpoints (/api/dashboard/ & /api/admin/):** GET /api/dashboard/stats.php computes rendered hours, target hours, progress percentage, remaining hours, and target countdown. Admin endpoints (GET /api/admin/dashboard/stats.php and GET /api/admin/students/list.php) provide cohort-wide aggregated metrics, real-time duty rosters, and section filtering (BSIT 4-A to 4-K).
- **Task Management Endpoints (/api/tasks/):** Provides full CRUD functionality (list.php, create.php, update_status.php), managing task titles, descriptions, deadlines, statuses (pending, in_progress, completed), and deliverable verification stamps.

4. Important Implementation Choices

- **Strict PDO Prepared Statements (Zero SQL Injection):** All database interactions use PHP Data Objects (PDO) with parameterized prepared statements. No raw user input is concatenated into SQL queries, completely eliminating SQL injection vulnerabilities and ensuring enterprise-grade data security.
- **Single Unified Sign-In with Intelligent Role Routing:** Rather than forcing users through separate, disjointed login screens, OnTrack implements a unified, modern sign-in interface. The backend automatically resolves the user's role (student intern vs. academic coordinator) from their credentials and routes them to their respective portal seamlessly.
- **Streamlined GPS-Centric Time-In with Realtime Clock:** Shifted from friction-heavy selfie capture to rapid GPS geolocation combined with a live system clock. This eliminates camera latency and bandwidth costs while ensuring verifiable location accuracy and tamper-proof temporal logging.
- **Database Performance Optimization via Composite Indexes:** Introduced strategic composite indexes (idx_user_date_time on attendance_logs and idx_user_id_desc on tasks). This resolves table scanning bottlenecks, guaranteeing sub-millisecond query responses even as attendance records grow over entire academic semesters.
- **Multi-Environment Network Base URL Configuration:** Structured environment.ts with configurable base API URLs, supporting rapid zero-reconfiguration switching between desktop browsers (localhost), Android Studio Emulators (10.0.2.2), and physical mobile hardware connected via local area network Wi-Fi.

PART 5: AI Usage Disclosure

1. Identification of AI Tools Used

- **Primary AI Tool:** Google Antigravity / Gemini AI Coding Assistant (DeepMind Advanced Agentic Environment).
- **Deployment Context:** Integrated agentic pair-programming assistant operating directly inside the codebase workspace.
- **Underlying Capabilities:** Context-aware static analysis, real-time TypeScript/PHP code synthesis, terminal automation, and native mobile build synchronization.

2. How the AI Assisted Development

- **Architecture & Scaffolding:** Assisted in structuring the Ionic 9 and Angular 22 standalone component architecture, configuring routes, models, and type-safe HTTP communication with the backend.
- **Device Hardware Integration:** Guided the configuration of Capacitor plugins (@capacitor/camera for deliverable photo capture, @capacitor/geolocation for workplace GPS coordinate recording, and @capacitor/haptics for tactile user feedback).
- **Backend RESTful API & Database Integration:** Constructed secure PHP REST API endpoints utilizing PDO prepared statements to eliminate SQL injection, implemented CORS headers, and authored the normalized MySQL database schema.
- **Native Android Bridge Configuration:** Generated and synchronized native Android platform assets, configured AndroidManifest.xml permissions (Camera, Location, and cleartext HTTP traffic), and prepared Gradle integration for Android Studio.
- **Bug Diagnosis & Query Optimization:** Diagnosed UI change detection locks during registration, applied Angular ChangeDetectorRef updates, and eliminated database latency by introducing targeted composite indexes on high-traffic tables.

3. What the Group Reviewed, Tested, or Modified

- **UI/UX & Mobile Visual Hierarchy Review:** The group reviewed mobile viewport layouts and removed redundant interface clutter—such as repeated company tags and unnecessary 'Off Duty' badges—to maintain a clean and accessible user experience.
- **Academic Workflow & Section Assignment Review:** Reviewed university coordination requirements and structured automated section mapping (BSIT 4-A through 4-K), ensuring newly registered students are automatically paired with their respective academic coordinators.
- **Database Schema & Integrity Verification:** Audited database tables and schema structures in phpMyAdmin to verify referential integrity, foreign key relations, and accurate storage across ontrack_db.
- **Comprehensive Functional Testing:** Conducted end-to-end testing of student registration, duplicate credential validation (handling HTTP 409 conflict states), GPS coordinate capture, deliverable proof uploads, and automated daily rendering hour calculation.

- **Multi-Environment Network Testing:** Tested API connectivity across web browsers (localhost), Android Studio Emulators (10.0.2.2), and physical mobile hardware connected via local Wi-Fi.
- **Administrative Scope Modifications:** Refactored the Coordinator Portal to eliminate unnecessary 'add task' buttons, refocusing the coordinator role purely on oversight, DTR auditing, and student deliverable verification.
- **Performance Tuning:** Added composite database indexes (idx_user_date_time and idx_user_id_desc), resolving slow loading times when fetching detailed student attendance histories.